

Putting Rational and Irrational Numbers in Order

Answer Key

Grade 8 / Pre-Algebra The number system

Quick Answers

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|---------------------------------------|---|
| 1. 7.07 | 7. — |
| 2. — | 8. 9.22 |
| 3. $\sqrt{10}$, $\frac{10}{3}$, 3.5 | 9. $3\sqrt{5}$, 6.7, $\frac{20}{3}$ |
| 4. 4.47 | 10. 4.25, — |
| 5. — | 11. $\frac{7}{5}$, 1.41, $\sqrt{2}$, 1.42 |
| 6. $\frac{27}{5}$, $\sqrt{30}$, 5.5 | 12. -2.3 , $-\frac{9}{4}$, $-\sqrt{5}$, 2.4, $\sqrt{6}$ |
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Curriculum

Level: Grade 8 / Pre-Algebra The number system

8.EE.A – *problems 1, 4, 8*

8.NS.A – *problems 2, 3, 5, 6, 7, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

4. *If they got 10: halved the radicand instead of taking its square root*
 8. *If they got 8.5: moved the decimal point one place instead of taking the square root*
 10. *If they got 17.4: read the fraction bar as a decimal point*
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1. $\sqrt{50}$ to the nearest hundredth.

Step 1. A root is placed by trapping it between perfect squares, so find the two that surround 50: $49 = 7^2$ and $64 = 8^2$.

Step 2. So $7 < \sqrt{50} < 8$, and since 50 is barely above 49 the value must sit just above 7.

Step 3. Refining, $7.07^2 = 49.98$ and $7.08^2 = 50.13$, so to the nearest hundredth $\sqrt{50} = \boxed{7.07}$.

Teaching note: the trapping step is what makes the calculator value checkable. A student who writes 25 here has not been stopped by anything.

2. $\sqrt{9}$ against 2.9.

Step 1. Not every square root is irrational — check for a perfect square before reaching for a decimal approximation.

Step 2. 9 is a perfect square, so $\sqrt{9} = 3$ exactly, and 3 against 2.9 is an ordinary decimal comparison.

Step 3. Three is greater, so $\sqrt{9} \boxed{>} 2.9$.

Listen for: “a square root is always irrational.” It is irrational only when the radicand is not a perfect square.

3. Order 3.5 , $\sqrt{10}$, $\frac{10}{3}$ from least to greatest.

Step 1. The three are written in three different notations, so nothing can be ordered until they share one — convert all to decimals.

Step 2. $\sqrt{10} \approx 3.162$ (since $3.16^2 = 9.99$), $\frac{10}{3} = 3.333\dots$, and 3.5 is already a decimal.

Step 3. Reading the decimals in order gives $\sqrt{10} < \frac{10}{3} < 3.5$.

Note: the answer is written with the original symbols, not the decimals — the decimals were the tool, not the answer.

4. $\sqrt{20}$ to the nearest hundredth.

Step 1. Trap it: $16 = 4^2$ and $25 = 5^2$ surround 20, so $4 < \sqrt{20} < 5$.

Step 2. 20 sits near the middle of that gap, so try 4.5: $4.5^2 = 20.25$, slightly too big. Try 4.47: $4.47^2 = 19.98$.

Step 3. To the nearest hundredth, $\sqrt{20} = \boxed{4.47}$, lying between 4 and 5.

Common wrong answer: 10 — the radicand was halved instead of square-rooted. Halving and square-rooting agree only at 4.

5. $\sqrt{2}$ against 1.5.

Step 1. Both values are positive, and squaring preserves order for positives, so squaring removes the radical without any estimation.

Step 2. $(\sqrt{2})^2 = 2$ and $1.5^2 = 2.25$. Since $2 < 2.25$, the original order is the same.

Step 3. So $\sqrt{2} < 1.5$.

Check: $\sqrt{2} \approx 1.414$, which is indeed below 1.5.

6. Order $\sqrt{30}$, 5.5 , $\frac{27}{5}$ from least to greatest.

Step 1. Convert to decimals, and keep three places — these three values differ only in the first two.

Step 2. $\frac{27}{5} = 5.4$; $\sqrt{30} \approx 5.477$ (since $5.48^2 = 30.03$); and 5.5 is given.

Step 3. In order: $\frac{27}{5} < \sqrt{30} < 5.5$.

Teaching note: the irrational value lands between the two rationals. That is the shape of most of these problems.

7. π against $\frac{22}{7}$.

Step 1. $\frac{22}{7}$ is a rational *approximation* of π , never equal to it, so the comparison turns on decimal places past the third.

Step 2. $\pi = 3.14159\dots$ and $\frac{22}{7} = 3.142857\dots$. They agree to two places and separate at the third.

Step 3. So $\pi < \frac{22}{7}$.

Listen for: “they are equal.” π is irrational, so no fraction of whole numbers can equal it — an equals sign here is always wrong.

8. $\sqrt{85}$ to the nearest hundredth.

Step 1. Trap it: $81 = 9^2$ and $100 = 10^2$, so $9 < \sqrt{85} < 10$.

Step 2. 85 is only 4 above 81 but 15 below 100, so the value is much nearer 9 than 10 — not half-way, because squaring stretches the gaps unevenly.

Step 3. $9.22^2 = 85.01$, so $\sqrt{85} = \boxed{9.22}$.

Common wrong answer: 8.5 — the decimal point was moved one place instead of taking a root.

9. Order $\sqrt{45}$, 6.7, $\frac{20}{3}$ from greatest to least.

Step 1. The direction is reversed here, so decide the decimal order first and reverse it once at the end rather than trying to think backwards throughout.

Step 2. $\frac{20}{3} = 6.667$; $\sqrt{45} \approx 6.708$ (since $6.71^2 = 45.02$); 6.7 is given. Least to greatest that is 6.667, 6.7, 6.708.

Step 3. Reversing, greatest to least is $\boxed{\sqrt{45} > 6.7 > \frac{20}{3}}$.

Common error: answering in the wrong direction. Two decimal places would also have made 6.7 and 6.708 look equal.

10. $\sqrt{18}$ against $\frac{17}{4}$.

Step 1 (a). Dividing is the conversion step: $17 \div 4$, so $\frac{17}{4} = \boxed{4.25}$ exactly.

Step 2 (b). Now estimate the root: $16 < 18 < 25$, so $\sqrt{18}$ is just above 4. More precisely $4.24^2 = 17.98$ and $4.25^2 = 18.0625$, so $\sqrt{18} \approx 4.243$.

Step 3 (b). Since $4.243 < 4.25$, the answer is $\sqrt{18} < \boxed{\frac{17}{4}}$.

Common wrong answer: 17.4 for part (a) — the fraction bar was read as a decimal point. Also note $4.25^2 = 18.0625$ is above 18, which by itself settles the comparison without any decimals.

11. Order $\sqrt{2}$, 1.42, $\frac{7}{5}$, 1.41 from least to greatest.

Step 1. All four values live inside a range of about 0.02, so two decimal places are not enough to separate them — carry at least four.

Step 2. $\frac{7}{5} = 1.4000$; 1.41 is 1.4100; $\sqrt{2} = 1.4142$; 1.42 is 1.4200.

Step 3. In order: $\boxed{\frac{7}{5} < 1.41 < \sqrt{2} < 1.42}$.

Teaching note: rounding $\sqrt{2}$ to 1.41 would have made two of these equal. How many places you carry is part of the method, not a detail.

12. Synthesis: order $-\sqrt{5}$, -2.3 , $-\frac{9}{4}$, $\sqrt{6}$, 2.4 .

Step 1. Split by sign first: three negatives and two positives, and every negative is less than every positive, so the two groups can be ordered separately.

Step 2. Negatives as decimals: -2.3 , $-\frac{9}{4} = -2.25$, $-\sqrt{5} \approx -2.236$. Further from zero means smaller, so the order runs -2.3 , then -2.25 , then -2.236 .

Step 3. Positives: 2.4 and $\sqrt{6} \approx 2.449$ (since $2.45^2 = 6.0025$), so 2.4 comes first.

Step 4. Putting the two groups together: $-2.3 < -\frac{9}{4} < -\sqrt{5} < 2.4 < \sqrt{6}$.

Listen for: -2.3 placed last among the negatives because “ 2.3 is the smallest number”. On the negative side the order reverses, and this is the one problem on the sheet that tests it.